UNIVERSITY OF SKÖVDE
SCHOOL OF INFORMATICS

# LINUX ADMINISTRATION 7.5 HP – IT387G

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# STUDENT MANUAL

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**Aim**

After completed course the student should be able to:

- Describe the fundamental mechanisms in Linux.
- Describe and discuss the area's development and the area's central concepts.
- Use tools such as schedulers and configuration management software to facilitate and automate administrative tasks.
- Install, configure, and document a Linux system.

The goals are examined with a practical lab assignment of 4 hp and a written exam of 3.5 hp.

**Requirements**

To achieve a pass in this assignment (which is required to pass the course), you must present your solution to one of the supervisors during a scheduled lab session and submit a lab report as a PDF via Canvas. You are expected to work independently and demonstrate a good level of initiative in finding the information needed to complete the assignment. Always attempt to solve problems yourself before seeking assistance. Supervisors expect you to have checked logs, manual pages, and documentation, and to have made a genuine effort to troubleshoot the issue before asking for help. Please note that Linux Administration at G1F level is a continuation course, so the expectations regarding your ability to independently locate information and resolve problems are higher than in previous G1N courses.

# Contents

# 1 INTRODUCTION (READ THIS CHAPTER FIRST)

Your task is to set up a Debian Linux server environment for a small company that requires its own Domain Name System (DNS), a mail server, and an accompanying web interface. If you encounter any unfamiliar concepts, consult the lecture material or research them online. Please note that this document constitutes the assignment itself and is not a step-by-step guide. The task is designed to encourage you, as a student, to independently seek out the necessary information to complete the assignment.

Throughout this document, you will find placeholders such as `[login]`, `[room]`, and `[group]`. These placeholders should be replaced with your university login, room number, and computer group number. For example, if your university login is a24login, and you are seated in room D209 at computer group 30, then `[login]` becomes a24login, `[room]` becomes 209, and `[group]` becomes 30.

## 1.1 Documentation

Do not forget to document your work thoroughly. Keep detailed notes and save configuration files in alternative locations, such as email, a USB stick, or a cloud storage service. Always make a copy of the original configuration file **before** making any changes. This practice makes it easier to track what modifications have been made. For example, using the copy command: `cp bind.named.local bind.named.local.orig` you made a backup of the config file, you can then use `diff -u bind.named.local.orig bind.named.local` to see what lines you have changed.

You can use an SSH file transfer client in the Netboot (e.g., SCP or SFTP) to transfer files from your virtual machines to the Netboot, allowing you to save them for future use. By the end of the lab, ensure that all necessary configuration files have been saved from your drives, as you may not retain access to the disks after the course.

## 1.2 Demonstration checklist

This is what you are going to show the supervisor when you are done:

1. Show how you can send an e-mail as `olivia.johnson@[login].it387g.nsa.his.se` via the webmail to `autoreply@nsa.his.se`.
2. In the webmail show the response from `autoreply@nsa.his.se` with the included fortune cookie to the supervisor.
3. Demonstrate that you can use passwordless authentication via SSH, from your management machine to your servers.

## 1.3 Report

This chapter will describe what the report must contain and the formal requirements on the report.

### 1.3.1 Contents

Instead of describing everything you have configured in the report, we want you to describe the full process behind the steps for the demonstration. So we want you to describe what services and protocols which will be used and needs to function. To be accepted the report must cover this:

- DNS records including MX and A records.
- Protocols such as SMTP, IMAP, SSH, and HTTP.
- Concepts including *mailbox*, *user*, *aliases*.

Someone without previous experience in setting up an email server should gain a full understanding of the basics of how an email reaches its destination and returns to the sender. Minor issues in the report may be tolerated, but always double-check all requirements before submitting. You can use RFC 1035 (DNS), RFC 3521 (SMTP), RFC 4253 (SSH) and RFC 9051 (IMAP)

In the report we also want you to include the public key for your account on the management machine. Make sure to store your private key, it can be useful to prove that you are the owner of the public key.

### 1.3.2 General report requirements

Submit your report via the assignment tool on Canvas.

There is no official or required template for the report. You may follow any reasonable layout, as long as you follow the formal requirements:

- Submit everything as a **single PDF** document.
- Filenames must be your username at the university followed by `.pdf` (e.g. `a24login.pdf`).
- Create a title page containing your full, legal name, student login, name of the course, and the submission date (report deadline).
- Use black text on white background except where colours are necessary (e.g. to highlight changes/differences and aid the reader).
- Paper size is A4.
- The text body's default font size is 11±1, and any computer output (e.g. configuration files/command outputs) is set in a monospaced font.
- The footer has to contain the actual page number and page 1 must start from the first page of content, which is usually the page with the introduction chapter.
- A table of contents is required, and you must have numbered sections and subsections (like the lab assignment).
- We recommend that you use flowcharts and diagrams to show information flow between services and user.
- If you need to include large sections of configuration, do so in appendices[1]. Use fixed-width font, no screenshots (nor photos), and consider syntax highlighting.
- Reference handling follows the APA or Harvard style.

[1] Information about what an appendix is can be found in Appendix A

More information about how to use proper citing and writing references can be found at the university's library at the links:

- https://www.his.se/biblioteket/skriva-och-referera/referera-kallor/
- https://www.his.se/en/library/write-and-cite/referencing/

All examinations in this course are individual. Sharing files with other students, whether through a web server or other means (e.g., Dropbox, USB drives, email, etc.), is considered cheating. Submissions that are (nearly) identical to other students' work or to materials found elsewhere (e.g., the Internet, books) may be regarded as cheating and are subject to disciplinary action. Furthermore, using generative AI tools such as ChatGPT, Copilot, or similar platforms to produce or assist in the creation of your work is considered cheating. More information on plagiarism and how to avoid it can be found on the course page and the University's website.

# 2 SETTING UP THE VIRTUALIZATION AND LAB ENVIRONMENT

The computer on the left will be your management machine, boot that one into the Netboot environment. This setup allows you to access documentation, visit Canvas (to read this assignment), copy files between systems, and establish SSH connections to the virtual machines that will be configured during the lab.

For virtualization of your lab systems, Proxmox VE will be installed on your right computer. Proxmox VE can be installed from the network boot menu that appears when the computer boots up, under Installers › Proxmox › Proxmox VE (make sure to select the latest stable version). Documentation for Proxmox VE server can be found at: https://pve.proxmox.com/pve-docs/.

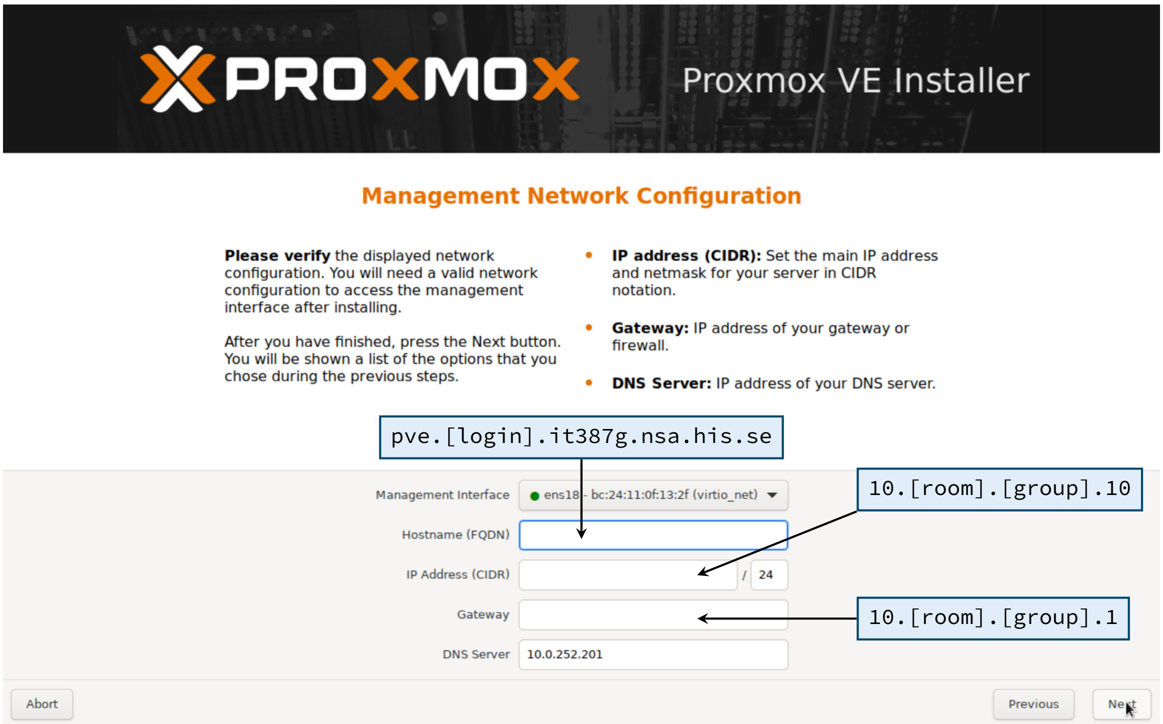
Username and password for the Proxmox server must be `root` and `Syp9393!`

## 2.1 Accessing and setting a new IP on the Proxmox host

During the installation of Proxmox you will be prompted with a few configuration questions. It is important that you select `zfs (RAID0)` under the *options* for *target harddisk* and configure proper FQDN and IPv4 addresses. For the latter, refer to Figure 1 – Proxmox VE network configuration.

**Figure 1**

*Proxmox VE network configuration*




You can use the console in Proxmox to manage virtual machines, but the most convenient way is to use the web interface. Access your Proxmox node from a web browser using the NSA Netboot client and type in your server's IP-address `https://10.[room].[group].10:8006`. The Proxmox server's IP address is displayed on the console screen of Proxmox after the installation is finished.

**Figure 2**

*Create VM*

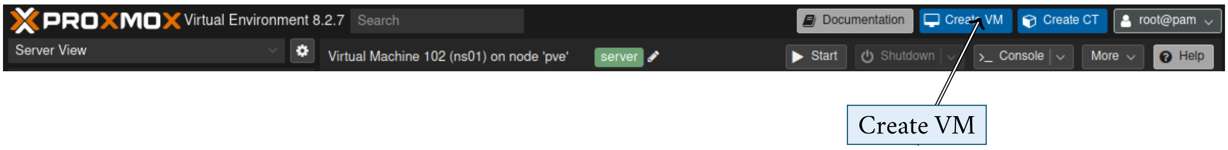



**Table 1**

*Specifications for the virtual machines*

| Name (also hostname) | IPv4 address | RAM | Hard drive |
|---|---|---|---|
| **ns1** | `10.[room].[group].11` | 2 GB | 20 GB |
| **ns2** | `10.[room].[group].12` | 2 GB | 20 GB |
| **mail** | `10.[room].[group].20` | 2 GB | 20 GB |
| **webmail** | `10.[room].[group].21` | 4 GB | 20 GB |

### 2.2 Setting up the Debian systems

Debian will be used on all of your virtual machines. Create virtual machines for all the servers using the '*Create VM*' option in the web interface (Figure 2).

Configure the machines with the following characteristics:

- Choose a name according to Table 1 and select a unique '*VM ID*'
- Set '*Guest OS*' to '*Linux*' and check the radio button for '*do not use any media*'
- Under '*System*', ensure that '*Default (SeaBIOS)*' is selected and that '*Qemu agent*' is enabled.
- For storage, select '*local-zfs*' and set the size according to Table 1.
  - If using an SSD, enable '*Discard*' and '*SSD emulation*'
- Under '*CPU*', set '*Type*' to '*host*' and assign the VM 2 cores.
- Under '*Memory*', allocate RAM to the VM according to Table 1
- Set '*Bridge*' to '*vmbr0*' under '*Network*'

> Some settings are under '*Advanced*' which you can show by enabling the check-box at the bottom of the window

Install the Debian 13 system using the NSA Netboot menu Installers > Linux > Debian 13 amd64.

When installing the newly created virtual machines, configure the systems according to the following specification:

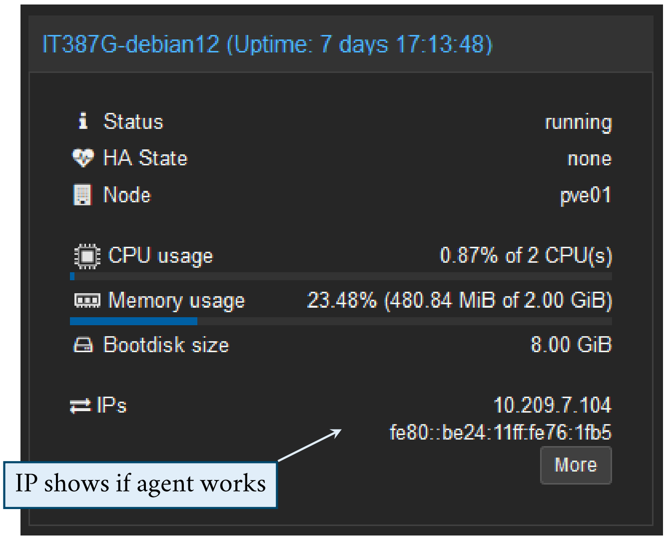
- They should not use any GUI.
- Language should be set to English with Brittish locale and Swedish keyboard.
- Hostname should follow the specifications in Table 1.
- Domain name must be `[login].it387g.nsa.his.se`, where `[login]` is replaced by your student university login.

- Use your university username as normal user account (e.g. *a24login*), all accounts must use the password **`Syp9393!`** (with exclamation mark '!').
- Software installation should be limited to the *SSH Server* and *Standard system utilities*.

Configure your systems to use `sudo` instead of `su`. Confirm that `sudo` is properly configured by running `sudo apt update` followed by `sudo apt upgrade` on every machine as normal user.

To ensure optimal performance in the virtualized environment, it is required to install the Qemu guest agent packages on your virtual machines. Debian provides a maintained package for this called `qemu-guest-agent`. If the VM was created correctly, this package is installed by default; however, if it is not, you should install it manually. After installation, you also need to alter VM options in Proxmox and enable *QEMU Guest Agent*. You need to power off/power on the virtual machine for this to take effect. With the agent installed your virtual machines will automatically power off nicely when you choose to shut down the virtualization environment.

**Figure 3**
*QEMU agent*




### 2.3 Administration of the systems

*Secure Shell* (SSH) is a vital protocol for administering Linux systems. Once the virtual machines are up and running, and the SSH daemon is configured along with the machine's IP address, **you must use an SSH client** (e.g., `ssh [username]@[IP]` from the terminal) to manage your systems. Using the Proxmox console is not considered good practice for system administration. The reasons are numerous, including the ability to have a resizable window when using the SSH terminal, as well as the capability to copy&paste text and transfer files via SSH.

Configure the SSH daemon to run on every system. Remote login as `root` using password is not allowed by default, you should SSH into your server as a normal user and then use `sudo` to invoke commands with root privileges (you will change this when you have done 3.1).

Editing configuration files is an essential task in many computing environments. To perform this, you will need to use a text editor. Popular choices include Vim and nano, each offering unique features and workflows. Selecting the right editor often depends on your preferences and the complexity of the task. Since Vim is included in the mandatory packages installed on all systems, a guide on its usage is provided in Appendix B. You are of course free to use nano if that is your cup of tea.

### 2.4 Log files

Use the `journalctl` command to see logs for specific services e.g. by using `journalctl -u bind9` for logs from the DNS service. Use `q` to quit from this mode.

If you want to see all log entries you can issue the `journalctl -f`, the command will not exit, instead it will show log entries as they are received. Use `Ctrl`+`c` to end.

`journalctl` writes the log files to the `/var/log/journal` directory. Some services have their own log files in `/var/log` directory, too.

From now and onwards in the lab your task is to always have a second SSH terminal session that shows the logs of the service you are currently configuring to quickly spot configuration issues by using the above commands. It is advantageous to arrange terminal windows next to each other.

> We recommend that you run `journalctl` with sudo or directly as root. It's possible to configure, so a normal user can read system logs – but it is not recommended.

### 2.5 Mandatory packages

Table 2 lists the mandatory packages required on all machines. Make sure these packages are installed; if any are missing, install them manually. These packages will be your very base installation of your Debian systems.

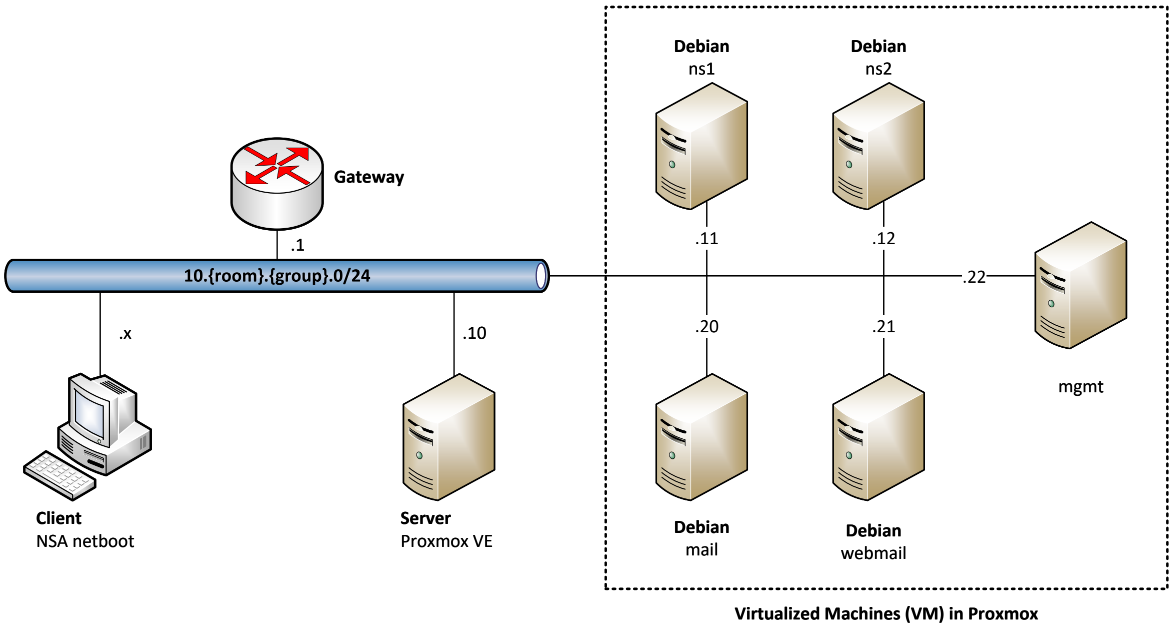
**Table 2**
*Mandatory packages*

| Package | Why it's needed |
|---|---|
| `vim` | Powerful file editor that Linux administrators use daily. |
| `qemu-guest-agent` | The virtual machine uses this to communicate with the virtualization platform to for example properly shutdown the guest. It also communicates the machine's IP address to Proxmox (see Figure 3). |
| `bind9-dnsutils` | To be able to troubleshoot DNS with the tools `dig` and `nslookup`. |

# 3 SETTING UP THE SERVICES

Your task is to set up the services according to Figure 4. It is your responsibility to find information for how to configure the services. To some services you may find examples in the lectures but also in the appendices and references that are available in this lab assignment.

**Figure 4**
*Topology*




## 3.1 Management machine (mgmt)

All proper security focused system administrators have a management machine that they use to maintain their server environment. As you have a high security focus, we want you to install a Linux Desktop machine that should be used to manage your servers. You are free to choose distribution and graphical environment for this machine. The only requirement on this machine is that:

- It must use the IP address 10.`[room]`.`[group]`.22/24.
- It must have an account using your username at the university (e.g. `[login]`).
- Password must be `Syp9393!`
- The `[login]` account must be prepared with SSH key pairs of type `ed25519` that will allow it to log in to all your other machines as root user.

From now on you can use this machine as a jump host https://en.wikipedia.org/wiki/Jump_server to access the other machines as root.

### 3.2 DNS & Redundancy (NS1 & NS2)

In order to get mail to work you must have a working DNS server in place with A and MX records. DNS is also useful to not have to memorize the IP addresses of the servers you are going to administrate. Please note that mail will not work if your DNS is not correctly configured.

> **You must use your allocated seat** otherwise the DNS delegation from our DNS servers will not work affecting your DNS and mail servers.

See Appendix C for the configuration **examples** of the master and slave DNS and also Zytrax online literature on Bind9 (ZyTrax, 2022)

Install two DNS servers using Bind 9: one will act as master and the other as slave. It is important that you use the hostnames and IP addresses as specified in Table 1. Ensure that all your virtual machines have static IP addresses with corresponding A records in the DNS, and that the MX records for your domain point to your mail server (see Appendix C). Configure the two name servers to allow queries for your domain from any source while restricting recursive queries to your own network. Test the configuration by querying the hosts in your domain using the `dig` command (see examples below). This must work both from the Netboot and from your internal machines, if it doesn't work from the Netboot there is something wrong with your DNS, or you are not seated in the correct computer group or (least likely) we messed up the DNS delegation.

Verify that others can find your mail server:

```
dig MX a24login.it387g.nsa.his.se
```

Query your DNS directly (asking for A records, recursive queries should only be possible from your own subnet):

```
dig @10.room.group.12 www.his.se
dig @10.room.group.11 mail.his.se
```

Ask the primary DNS in the NSA lab what it knows about your secondary DNS:

```
dig @10.0.252.201 ns2.login.it387g.nsa.his.se
```

To avoid caching issues and save time on troubleshooting during the server configuration, it is advisable to set the `$TTL` option to a low value, such as `30`, when testing your DNS. After your DNS servers are working correctly, configure all Debian machines to use both your DNS servers as their sole DNS service by updating their `/etc/resolv.conf` files.

> **Needed service packages:** `bind9`

**Table 3**
*Employees of your company*

| Full name | Username |
|---|---|
| Liam Smith | `lism` |
| Olivia Johnson | `oljo` |
| Ethan Brown | `etbr` |
| Mia Davis | `mida` |
| Lucas Miller | `lumi` |

### 3.3 Mail server

Your next task is to install an mail server using Postfix as SMTP server (Simple Mail Transfer Protocol) and Dovecot for IMAP (Internet Message Access Protocol). Use the machine specified as '*mail*' in Table 1 for this task. Since the mail server is only accessible within the organization, there is no need to install SSL/TLS encryption on the IMAP or SMTP server.

Your server should only handle the following two types of e-mails:

1. From any other server to your own domain, `[login].it387g.nsa.his.se`, as the final destination.
2. From your own (virtual) computers to any other destination, but only if the sender authenticated itself towards your SMTP server when submitting the mail.

The SMTP server must be configured to receive e-mails for the users listed in Table 3. These users must exist as accounts on the e-mail server.

Users should have full e-mail addresses, e.g., `olivia.johnson@[login].it387g.nsa.his.se`, which the mail server must map to local user accounts, e.g., `oljo`, for both outgoing and incoming emails. The process of mapping account names to full e-mail address can be solved using '*aliases*'. There is *no requirement* that users must be able to log in with their full e-mail address.

Test your configuration with telnet according to Ubuntuwiki (2013) and make sure that your SMTP server does not act as an open relay.

You can use Mozilla Thunderbird to test your mail configuration by sending an e-mail to `autoreply@nsa.his.se` from one of your accounts.

Make sure to verify that everything so far works as intended. There is no point trying to start with the next step before you have done this.

**Needed service packages:** `postfix`, `dovecot-core`, `dovecot-imapd`

### 3.4 Web mail

Your final task is to install and add a webmail interface to your e-mail solution. It must be installed on a dedicated machine. It is important that you have verified that your e-mail services is working before continuation with this step.

The application you are going to use for this is Roundcube https://roundcube.net/. It is a web interface implementation of an email client which is running on PHP https://www.php.net/. Debian is shipping with a stable version of Roundcube, but before installing that package it is important that we have a MySQL compatible SQL server to store all user settings in. So start of with installing `mariadb-server` before installing Roundcube (otherwise you will end up with problems). The Roundcube package in Debian will also install Apache HTTP Server and PHP support

You must configure Roundcube to use your IMAP and SMTP server to present and send e-mails. When a user accesses your `http://webmail.[a24login].it387g.nsa.his.se` they must end up in Roundcube (via a redirection or other solution).

**Needed service packages:** `mariadb-server`, `roundcube`

## References

Authors, A. w. c. (2026). Archlinux wiki - tmux [https://wiki.archlinux.org/title/Tmux [Accessed: 2026-01-09]].

Ubuntuwiki. (2013). Smtp, testing via telnet [http://ubuntuwiki.net/index.php/SMTP,_testing_via_Telnet [Accessed: 2025-12-01]].

ZyTrax. (2022). Dns for rocket scientists [https://www.zytrax.com/books/dns/ [Accessed: 2025-12-01]].

# Appendix A WHAT IS AN APPENDIX?

An appendix in a report or set of instructions is a supplementary section located at the end of the document. Its primary purpose is to provide additional information, resources, or supporting materials that are relevant to the content but not essential or extensive to be included in the main body of the document. By placing this information in the appendix, the report remains concise and easy to read, while still offering access to detailed or technical information for readers who may need it.

The types of content included in an appendix can vary widely depending on the nature of the document. Examples include raw data, detailed calculations, supplementary graphs, charts, or images, as well as transcripts, survey questions, glossaries, or technical specifications. For instructional documents, appendices might contain step-by-step guides, configuration files, or example full code blocks.

Appendices are labelled using capital letters instead of numbers (e.g., *Appendix A*, *Appendix B*, etc.) and are often organized in the order they are referenced in the main text. When referring to an appendix, the main content should clearly direct readers to the relevant section, for instance, by stating, "See Appendix A for detailed survey results." This structure ensures that the document is both thorough and user-friendly, catering to a range of readers with varying levels of interest or expertise in the topic.

**Example of configuration in an appendix**

```
// Bind9 named.conf.local example
zone "example.com" {
    type master;
    file "/etc/bind/db.example.com";
};

zone "1.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.1";
};
```

In this configuration, the `zone "example.com"` block defines a forward zone for the `example.com` domain. The `zone "1.168.192.in-addr.arpa"` block defines a reverse DNS zone for the IP range `192.168.1.x`, and zone files are stored in the `/etc/bind/` directory.

# Appendix B VIM COMMANDS

**Table 4**
*Most Used Vim Commands – highly simplified*

| Category | Command and Description |
| --- | --- |
| **Basic Navigation** | |
| `h, j, k, l` | Move left, down, up, right (arrow keys usually works) |
| `w, e, b` | Move by word forward, to end, backward |
| `0, ^, $` | Start of line, first non-whitespace, end of line |
| `gg, G` | Move to start or end of file |
| `5G` | Jump to line 5. 1 first line, 0 is end of file |
| **Insert Mode** | |
| `i` | Insert before cursor |
| `a` | Append after cursor |
| `o, O` | Open a new line below or above |
| Esc | Exit insert mode |
| **Saving/Exiting** | |
| `:w` | Save changes |
| `:q` | Quit |
| `:wq` / `ZZ` | Save and quit |
| `:q!` | Quit without saving |
| **Deleting/Copying** | |
| `x` | Delete character under cursor |
| `dd` | Delete the current line |
| `yy` | Yank (copy) the current line |
| `p, P` | Paste after or before cursor |
| **Searching** | |
| `/pattern` | Search forward for a pattern |
| `?pattern` | Search backward for a pattern |
| `n, N` | Repeat search forward or backward |
| **Undo/Redo** | |
| `u` | Undo the last action |
| Ctrl + r | Redo the undone action |
| **Other Useful** | |
| `:set nu` | Show line numbers |
| `:%s/old/new/g` | Replace `old` with `new` globally |
| `v` | Visual select |

Some commands can also be used together with quantifiers, such as `dd` where you, e.g., can use the command `5dd` to delete five lines instead of just one.

It's important to understand that Vim is using different modes, there is `Normal mode`, `Insert mode`, `Command mode`, `Visual mode` and `Replace mode`. For more information about vim, check the documentation: https://www.vim.org/docs.php

## Appendix C BIND9 EXAMPLE CONFIGURATION

> Names and IP must be changed to fit your environment.

Remember to always check the default files already existing and update them where necessary. There is no need to delete any default files, only make sure to edit them.

### `/etc/bind/named.conf.options`

Your nameservers have to forward all queries they cannot answer to the lab servers 10.0.252.201 and 10.0.252.202. We allow anyone to ask queries about the domains we are authoritative for, but only localhost and local networks may use us as their main resolver.

```
options {
    directory "/var/cache/bind";
    # Recursive questions get sent to NSA nameservers
    forwarders {
        10.0.252.201;
        10.0.252.202;
    };

    dnssec-validation auto;

    auth-nxdomain no; # conform to RFC1035
    listen-on-v6 { any; };

    allow-query { any; };
    allow-recursion { 127/8; 10.room.group.0/24; };
};
```

### `/etc/bind/named.conf.local` (on DNS master ns1)

We are the authoritative name server for the DNS zone `[login].it387g.nsa.his.se` so we include its zone file so that the bind9 daemon will read it. A proper installation would also include a reverse zone. This example also set to allow zone transfers to the `ns2` slave DNS server.

```
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "group.room.10.in-addr.arpa" {
    type master;
    file "/etc/bind/db.10";
```

```
    allow-transfer { 10.room.group.12; };
};

zone "login.it387g.nsa.his.se" {
    type master;
    file "/etc/bind/db.login";
    allow-transfer { 10.room.group.12; };
};
```

The below examples are not complete, and it is up to you to identify the proper entries needed.

#### `/etc/bind/db.[login]` (on DNS master ns1)

To better understand the SOA record at the beginning of the zone file, please read http://www.zytrax.com/books/dns/ch8/soa.html. This is a minimal example, see http://www.zytrax.com/books/dns/ch8/#types for more possible record entries.

```
$TTL 300
$ORIGIN login.it387g.nsa.his.se.
@             IN        SOA ns1 root.login.it387g.nsa.his.se. (
                                 2025121301    ; Serial
                                 30m           ; Refresh
                                 3m            ; Retry
                                 2w            ; Expire
                                 1h )          ; Negative Cache TTL
              IN        NS       ns1
              IN        NS       ns2
@               IN        MX       10     mail

; Our A records mapping host to IP
ns1           IN        A            10.room.group.11
...           ...       ...          ...
...           ...       ...          ...
...           ...       ...          ...
mgmt       IN        A            10.room.group.22
```

#### `/etc/bind/db.10` (on DNS master ns1)

```
$TTL 300
@          IN      SOA ns1 root.login.it387g.nsa.his.se. (
                                 2025121201    ; Serial
                                 30m           ; Refresh
                                 3m            ; Retry
                                 2w            ; Expire
                                 1h )          ; Negative Cache TTL
@          IN      NS              ns1.login.it387g.nsa.his.se.
@          IN      NS              ns2.login.it387g.nsa.his.se.
```

```
; Our reverse mapping IP to host
11        IN      PTR            ns1.login.it387g.nsa.his.se.
...       ...     ...            ...
...       ...     ...            ...
...       ...     ...            ...
22        IN      PTR            mgmt.login.it387g.nsa.his.se.
```

#### `/etc/bind/named.conf.local` (on DNS slave ns2)

On the slave, we only refer to the master, we do not need to maintain databases here.

```
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "group.room.10.in-addr.arpa" {
    type slave;
    masters { 10.room.group.11; };
};

zone "login.it387g.nsa.his.se" {
    type slave;
    masters { 10.room.group.11; };
};
```

# Appendix D READING LOGS — HINTS AND TIPS

This appendix will give you some good tips on how to interpret log output from `journalctl`.

In the following example we will see a log entry produced by Google Chrome, it follows some patterns that are good to learn.

```
jan 04 18:30:28 aluminium google-chrome.desktop[21513]:
  [21566:21651:0104/183028.651622:ERROR:components/viz/service/display/display.cc:277] Frame
  latency is negative: -0.134 ms
```

We can read out the following:

- Time when it happened `jan 04 18:30:28` and as we don't get a timezone, we assume it is the local timezone.
- Hostname it happened on was `aluminium`
- Program is defined in `google-chrome.desktop`
- Process ID of the program is/was `21513`. Thus, we can use *ps* to find out if the process is still running and how it was started *ps -auwq 21513*
- We also get a filename, in this case it probably refers to the line in the original source code where the message was produced. In this case in file `components/viz/service/display/display.cc` at line number `277`. The pattern with a filename followed by colon and a number usually refers to a specific line in a file, very common to see this kind of output when we have made a syntax error in a configuration file.

Another example where we have syntax error in a configuration file:

```
root@ns1:~# journalctl -u bind9 -n 8
-- Logs begin at Thu 2025-11-27 15:16:45 CET, end at Mon 2026-01-05 14:38:17 CET. --
Jan 05 14:38:17 ns1 named[636]: configuring command channel from '/etc/bind/rndc.key'
Jan 05 14:38:17 ns1 named[636]: reloading configuration succeeded
Jan 05 14:38:17 ns1 named[636]: reloading zones succeeded
Jan 05 14:38:17 ns1 named[636]: all zones loaded
Jan 05 14:38:17 ns1 named[636]: running
Jan 05 14:38:17 ns1 named[636]: dns_rdata_fromtext: /etc/bind/db.b10johza:17: near '.': extra
  input text
Jan 05 14:38:17 ns1 named[636]: zone caro.it387g.nsa.his.se/IN: loading from master file
  /etc/bind/db.b10johza failed: extra input text
Jan 05 14:38:17 ns1 named[636]: zone caro.it387g.nsa.his.se/IN: not loaded due to errors.
```

We can see:

- We are looking at the eight last lines from system service (unit) bind9, that is `named` the DNS server daemon.
- That Bind is running with process id 636.
- We see a familiar pattern `/etc/bind/db.b10johza:17: near '.': extra input text`, filename with a line number. So on line 17 in `/etc/bind/db.b10johza` it complains about something near a dot. Next lines in the output says it failed to load the zone data because of this syntax error.

### Outside systemd

Some applications have not fully migrated to use `journalctl` as it logging backend. One area where this is the case is with web applications, which can be hard enough to debug without this extra level of complexity. Apache and PHP applications is among those. So even if there is a `journalctl` log for `apache2` you will find out it is mostly empty, at least it doesn't contain any valuable debug information. Apache logs[2] to text files under `/var/log/apache2`. They are by default split up to `access.log` which records every page request and `error.log` which contains errors from Apache2. So you would maybe expect to find errors from PHP applications there (as they run under Apache2), traditionally you did, but nowadays PHP applications mostly log to their dedicated files. So for example Roundcube which is implemented in PHP sends it logs to the folder `/var/log/roundcube`.

Here is an extract from `/var/log/roundcube/error.log` where oljo fails to login to Roundcube:

```
[01-Dec-2025 13:19:21 +0000]: <f800fdc1> IMAP Error: Login failed for oljo against ↙
  mail.caro.it387g.nsa.his.se from 10.0.102.131. AUTHENTICATE PLAIN: Authentication failed. ↙
  in /usr/share/roundcube/program/lib/Roundcube/rcube_imap.php on line 211 (POST ↙
  /roundcube/?_task=login&_action=login)
```

[2]With a default configuration, you can change this.

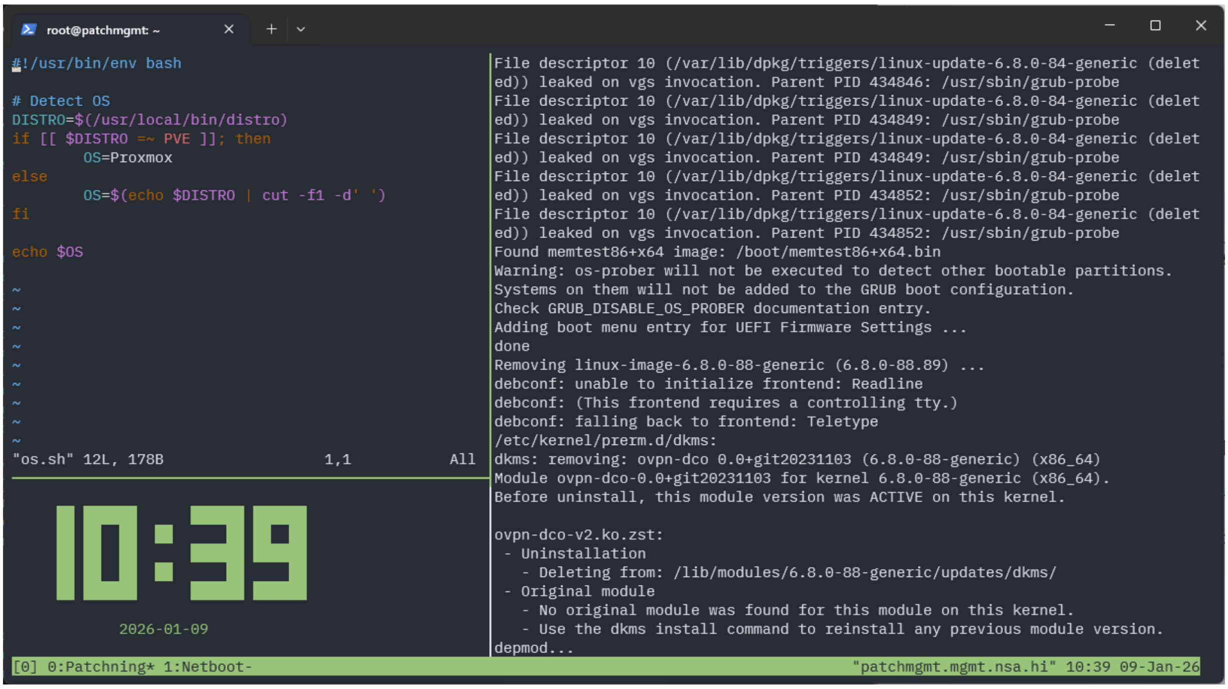
# Appendix E QUICK INTRODUCTION TO TMUX

What is `tmux`? `tmux` is a terminal multiplexer, it enables a number of terminals (or panes), each running a separate program or shell, to be created, accessed, and controlled from a single terminal (see Figure 5). tmux may be detached from a screen and continue running in the background, then later reattached (Authors, 2026).

With `tmux` you don't need to use the mouse to position windows on the screen, and once you get used to it you can work way faster and access information more conveniently. If you are going to do a lengthy operation over SSH `tmux` is a must. Normally if you get disconnected running SSH, whatever you were doing on the server would be killed, but with `tmux` you can simply reconnect to the server and reattach to the `tmux` session. Not only can you reconnect, but you can have several connections to the same `tmux` session - this means you can be connected to the same session locally on a server console and remotely via SSH[3].

**Figure 5**

*tmux with two windows. Showing window "Patchning" which has screen split in three panes; running tty-clock, vim and apt-get.*




To send commands to `tmux` itself you type a hotkey, using the default configuration this hotkey is Ctrl + b. So you first enter the hotkey, then the command to `tmux`. So pressing Ctrl + b, releasing, then pressing % will split the current pane in half vertically. See Table 5 for keyboard commands.

[3]As the number of rows and columns can differ between clients, `tmux` automatically adapts screen layout to the attached client with the lowest resolution

**Table 5**
*Default tmux shortcuts.*

| Shortcut | Description |
|---|---|
| **Basic pane handling** | |
| Ctrl + b ↑ ↓ ← → | Navigate between panes. |
| Ctrl + b % | Split current pane in half vertically. |
| Ctrl + b " | Split current pane in half horizontally. |
| Ctrl + b pg up | Halt output, scroll back in pane buffer – end with q. |
| Ctrl + d or `logout` | End current pane. If last, ends tmux. |
| Ctrl + b x | Kill current pane.<br>If last pane and more windows, closes current window.<br>If last pane and window, closes tmux. |
| Ctrl + b z | Zoom current pane to fullscreen, if zoomed exit zoom. |
| Ctrl + b Ctrl + arrow | Resizes current pane in arrow direction. |
| **Using windows** | |
| Ctrl + b c | Create new window. |
| Ctrl + b n | Change to next window in list. |
| Ctrl + b p | Change to previous window in list. |
| Ctrl + b l | Change to previously active window. |
| Ctrl + b , | Rename active window. |
| Ctrl + b 0-9 | Change to numbered window. |
| **Session and other** | |
| Ctrl + b d | Detach from session. |
| `tmux attach` | To reconnect again. |
| Ctrl + b D | Detach another session. |
| Ctrl + b r | Redraw screen. |
| Ctrl + b t | Show a clock. |
| Ctrl + b ? | List of all shortcuts. |
| **Text Commands** | |
| Ctrl + b : | Enter tmux interactive command prompt. |
| `set mouse off` | Turn off tmux mouse support |
| `set mouse on` | Turn on tmux mouse support |